

Practical Session 13: Evaluating LLMs Honestly

Lecture 13 — Calibration, Contamination Audits, and Trading-Grade Metrics

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Session Overview

Duration: ~1 hour (50 min hands-on + 10 min debrief)

Timed agenda

- 1 **Recap** (5 min): ECE, temporal splits, Sharpe/IR, transaction costs.
- 2 **Problem 1** (12 min): Compute ECE & MCE by hand from bin statistics.
- 3 **Problem 2** (12 min): Design a contamination-resistant split; quantify a leakage audit.
- 4 **Problem 3** (10 min): Gross vs. net Sharpe under transaction costs.
- 5 **Case study** (15 min): Python — reliability diagram + temperature scaling on a toy LLM credit classifier (code/notebooks/13-llm-limitations-evaluation/).
- 6 **Discussion** (6 min): Would you deploy this model?

Recap: The Four Formulas You Need Today

Expected / Maximum Calibration Error:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \bar{p}(B_m)|, \quad \text{MCE} = \max_m |\text{acc}(B_m) - \bar{p}(B_m)|.$$

Temporal split with gap $\Delta \geq h$:

$$\text{train} : t \leq T_1 - \Delta, \quad \text{valid} : T_1 \leq t \leq T_2 - \Delta, \quad \text{test} : t \geq T_2.$$

Sharpe and net-of-cost return:

$$\text{SR} = \frac{\bar{R}_p - R_f}{\sigma_p}, \quad R_p^{\text{net}} = R_p^{\text{gross}} - 2\tau(\text{half-spread} + \text{impact}).$$

Recall

Temperature scaling $\text{softmax}(\mathbf{z}/T)$ with $T > 1$ cures overconfidence and *preserves ranking* (AUROC unchanged).

Problem 1: Compute ECE and MCE

An LLM sentiment model labels 1,000 headlines as “price-positive”. Confidence binned into 4 bins:

Bin	Interval	$ B_m $	$\bar{p}(B_m)$	$\text{acc}(B_m)$
1	[0.0, 0.25)	100	0.15	0.18
2	[0.25, 0.5)	200	0.40	0.35
3	[0.5, 0.75)	300	0.65	0.50
4	[0.75, 1.0]	400	0.90	0.66

Tasks.

- 1 Compute the per-bin gap $|\text{acc} - \bar{p}|$.
- 2 Compute **ECE** (weight by $|B_m|/n$) and **MCE**.
- 3 In which confidence region is the model worst? What does that imply for a desk that only acts on > 0.75 signals?

Solution: Problem 1

Per-bin gaps: 0.03, 0.05, 0.15, 0.24. **Weights:** 0.10, 0.20, 0.30, 0.40.

$$\begin{aligned}\text{ECE} &= 0.10(0.03) + 0.20(0.05) + 0.30(0.15) + 0.40(0.24) \\ &= 0.003 + 0.010 + 0.045 + 0.096 = \mathbf{0.154}.\end{aligned}$$

$$\text{MCE} = \max\{0.03, 0.05, 0.15, 0.24\} = \mathbf{0.24}.$$

Interpretation

ECE = 15.4%, MCE = 24%, both in the top bin. The desk acting only on > 0.75 signals is operating in the model's **worst** region: stated 90% confidence corresponds to **66%** realised accuracy. Recalibrate (or abstain) before trusting high-confidence calls.

Problem 2: Design a Contamination Audit

A team evaluates an LLM (training cutoff **Dec 2023**) on predicting 1-month stock direction from earnings calls. Prediction horizon $h = 21$ trading days.

Tasks.

- 1 Give concrete cutoffs T_1, T_2 and gap Δ for a temporal split that is safe against this model. Justify Δ .
- 2 The team measures accuracy on *original* (62%) vs. *anonymised* (54%) transcripts. Quantify the leakage and state what it implies about the true edge.
- 3 Name two further controls and what each detects.

Solution: Problem 2

(1) Split. All test data must post-date the cutoff plus a publication-lag margin. Use $T_2 = \text{Apr 2024}$ (Dec 2023 cutoff + ≥ 3 months); T_1 in early 2023 for validation.

$\Delta \geq h + \text{lag} \approx 21\text{d} + \text{a few weeks} \Rightarrow \text{take } \Delta = 1 \text{ month so labels cannot straddle the boundary.}$

(2) Leakage. $62\% - 54\% = 8$ pp of accuracy was entity-level memorisation. Residual 54% still beats 50% but the *apparent* edge shrank by more than half—likely economically negligible after costs.

(3) Two further controls.

- **N-gram dedup** (13-grams, MinHash LSH) — detects direct memorisation of the document text.
- **Date-blinding ablation** — a large drop when temporal markers are removed reveals temporal pattern-retrieval rather than reasoning.

Problem 3: Gross vs. Net Sharpe

A daily news-sentiment LLM strategy reports, before costs:

$$\overline{R}_p^{\text{gross}} - R_f = 12\% \text{ annual}, \quad \sigma_p = 15\% \text{ annual}.$$

It rebalances daily with turnover $\tau = 0.5/\text{day}$; half-spread = 2 bps, market impact = 8 bps.

Tasks.

- 1 Compute the **gross** annual Sharpe ratio.
- 2 Daily round-trip cost = $2\tau(\text{half-spread} + \text{impact})$. Annualise over 252 trading days.
- 3 Compute the **net** Sharpe (subtract annual cost from excess return; σ_p unchanged). Is the strategy still worth trading?

Solution: Problem 3

(1) **Gross Sharpe:** $SR_{\text{gross}} = 12/15 = 0.80$.

(2) **Cost.** Daily round-trip = $2(0.5)(2 + 8)$ bps = 10 bps/day. Over 252 days:
 $10 \text{ bps} \times 252 = 2520 \text{ bps} = 25.2\%/\text{year}$.

(3) **Net Sharpe.**

$$\overline{R}_p^{\text{net}} - R_f = 12\% - 25.2\% = -13.2\%, \quad SR_{\text{net}} = \frac{-13.2}{15} = -0.88.$$

Verdict

A respectable-looking gross Sharpe of 0.80 turns **deeply negative** once daily-rebalancing costs are charged. The “edge” was an artefact of ignoring transaction costs—exactly the artefact the lecture warns about. Lower turnover or longer holding periods are mandatory before this signal is tradeable.

Case Study: Reliability + Temperature Scaling

Goal. Reproduce the credit-classifier reliability diagram, compute its ECE, then apply temperature scaling and watch ECE fall while AUROC is unchanged.

Reference code: `code/notebooks/13-llm-limitations-evaluation/`

- `gen_reliability_diagram.py` — deterministic diagram from the fixed bin statistics ($\text{ECE} = 0.115$).
- `exercises.ipynb` / `demo.ipynb` — calibration experiments.

What you will observe

- 1 The high-confidence bins sit *below* the diagonal — overconfidence.
- 2 Fitting $T > 1$ spreads the softmax: bins move toward the diagonal, ECE drops.
- 3 The ROC curve (and AUROC) is **unchanged** — scaling is rank-preserving. Calibration $\neq$ discrimination.

Case Study: The Code (1/2) — ECE

```
import numpy as np

# Fixed bin stats from the credit example (ex:ch13-ece-credit).
conf = np.array([0.08, 0.31, 0.50, 0.71, 0.88]) # mean predicted prob
acc = np.array([0.11, 0.28, 0.47, 0.55, 0.62]) # empirical accuracy
count = np.array([120, 180, 200, 300, 200])

def ece(conf, acc, count):
    w = count / count.sum()
    return float(np.sum(w * np.abs(conf - acc)))

print("ECE =", round(ece(conf, acc, count), 3)) # -> 0.115
print("MCE =", round(float(np.max(np.abs(conf - acc))), 3))
```

Case Study: The Code (2/2) — Temperature Scaling

```
# Temperature scaling on raw logits z: p = softmax(z / T).  
# T > 1 spreads the distribution and reduces overconfidence;  
# because it is monotone in z, the ranking (AUROC) is preserved.  
def temperature_scale(logits, T):  
    z = logits / T  
    z = z - z.max(axis=1, keepdims=True)          # numerical stability  
    e = np.exp(z)  
    return e / e.sum(axis=1, keepdims=True)
```

Case Study: Diagnostic Questions

Run `gen_reliability_diagram.py`, then answer:

Q1. Which bins lie *below* the diagonal, and what does that say about the model's confidence in that region?

Q2. The reported ECE is 0.115. Recompute it using only bins 4 and 5. Why do these two bins dominate the total?

Q3. You fit $T = 1.6$ and ECE falls to ≈ 0.04 , but AUROC is unchanged at 0.78. Explain in one sentence why calibration improved while discrimination did not.

Q4. A colleague proposes raising the action threshold from 0.5 to 0.8 instead of recalibrating. Does that fix the calibration problem? What does it fix, and what does it not?

Discussion: Would You Deploy This Model?

You are on a model-risk committee (SR 11-7). An LLM credit model shows: $ECE = 0.115$, 8 pp of its accuracy attributable to entity memorisation, and a news-overlay trading sleeve with gross Sharpe 0.80 / net Sharpe -0.88 .

Debate, framed around

- **Calibration** — deploy with recalibration + conformal sets, or not?
- **Contamination** — is the residual 54% edge real or an artefact?
- **Economics** — can lower turnover rescue the trading sleeve?
- **Governance** — what monitoring cadence (quarterly recalibration?) and human-in-the-loop controls would you require before sign-off?

Takeaway: LLM outputs are *inputs to a decision process*, not end products. The committee's job is the surrounding evaluation discipline.

APPENDIX

Stretch Problems

Extra / optional material — beyond the core lecture.

Stretch: Information Ratio Decomposition

A long-short LLM signal covers $N = 100$ independent monthly bets with Information Coefficient $IC = 0.04$.

Tasks.

- 1 Estimate the Information Ratio via $IR \approx IC\sqrt{N}$.
- 2 Is this “strong” ($IR > 0.5$) or “exceptional” ($IR > 1.0$)?
- 3 To reach $IR = 0.8$ at the same IC , how many bets N would you need? What practical cost does increasing N introduce?

Solution sketch. $IR \approx 0.04\sqrt{100} = 0.40$ (below “strong”). For $IR = 0.8$:
 $\sqrt{N} = 0.8/0.04 = 20 \Rightarrow N = 400$ bets/year. More bets means higher turnover $\Rightarrow$ higher transaction costs, which can erase the gain (see Problem 3).

Stretch: Walk-Forward with a Purge Period

You backtest a monthly-rebalanced LLM signal ($h = 1$ month) over 2019–2024.

Tasks.

- 1 Sketch the train/eval windows for the first three walk-forward iterations with purge $\Delta_{\text{purge}} = 1$ month.
- 2 Why must $\Delta_{\text{purge}} \geq h$? What leaks if it is zero?
- 3 For full LLM fine-tuning (expensive to re-estimate), what is the cost of widening h to amortise re-training?

Solution sketch. Iteration k : train $t \leq kh - \Delta_{\text{purge}}$, eval $kh \leq t \leq (k+1)h - 1$. With $\Delta_{\text{purge}} < h$, a label window overlapping the boundary appears in both sets $\Rightarrow$ look-ahead leakage. Widening h amortises re-fitting cost but evaluates an increasingly *stale* model late in each window.